Differentiable Quantum Architecture Co-Search for Reinforcement Learning

Jayden John
Department of Computer Science, University of Maryland, College Park, MD, USA
Email: jjohn111@umd.edu

***Abstract*—Variational quantum circuits (VQCs) offer compact, trainable models that may reduce parameter counts and improve stability for reinforcement learning (RL) in the noisy intermediate-scale quantum (NISQ) era, but their performance depends strongly on design choices in the encoding, ansatz, and readout stages. This paper proposes Differentiable Quantum Architecture Co-Search (DQACS), a fully differentiable framework that jointly optimizes variational gate placements together with trainable encoding maps and readout layers. DQACS is integrated into a hybrid quantum–classical Proximal Policy Optimization (PPO) pipeline with framestacking and a top-k architecture selection schedule to improve training stability. Experiments on MiniGrid and FrozenLake tasks show that the method can learn diverse environments and that convergence improves after the search collapses to a high-probability architecture.**

***Index Terms*—quantum reinforcement learning, variational quantum circuits, differentiable architecture search, PPO, NISQ**

I. Introduction

Quantum computing has made many significant advancements in the past couple of years with many new subfields of quantum computing being created. Despite this, in the current Noisy Intermediate Scale Quantum era (NISQ) we are limited by the number of qubits and the noise, that is errors that accumulate in long circuits. This is why for machine learning tasks with a quantum computer Variational Quantum Circuits (VQCs) are used [1]. These are quantum circuits with trainable parameters for the quantum gates allowing it to mimic a classical dense layer. The difference is that VQCs do not need as many trainable parameters as a dense layer. The benefits of this include reduced training times, as fewer parameters need to be calculated during backpropagation [2]. It also improves stability, as there are fewer parameters to optimize and the weight takes much less memory to store. It consists of an encoding layer which turns classical data into quantum data, the ansatz layer which is the trainable quantum circuit and the readout layer which converts the quantum output to classical data. By combining quantum hardware with Machine learning frameworks, Quantum reinforcement learning emerged with the goal of one day beating classical (non-quantum) reinforcement learning. Traditional classical reinforcement learning often requires heavy duty deep neural networks to navigate complex tasks which can be computationally intensive and involve a lot of parameters. VQCs do not have this issue as they are really compact. The downside of this is that it is hard to design as its accuracy is strongly tied to the arrangement of encoding layers, the ansatz structure, and the readout strategy. For reinforcement learning where it has to learn a variety of very different tasks it is infeasible to design all of the circuits by hand. The solution to this is Differentiable Quantum Architecture Search [3] which allows us to generate the gates of the quantum circuit itself which solves the design problem. However this only optimizes the ansatz, leaving a crucial need for an approach that optimizes the entire pipeline. This is why I propose differentiable quantum architecture Co-Search, a framework that fully optimizes the VQC. Building upon the foundations of DQAS, my approach implements a joint optimization strategy that encompasses not only the variational gate placements but also the trainable encoding maps and readout layers. By integrating this co-search mechanism within a hybrid quantum-classical Proximal Policy Optimization (PPO) framework and utilizing techniques such as framestacking, I aim to improve training stability and performance across complex environments. Throughout this work I aim to prove that my unified, differentiable search over the entire quantum circuit architecture will achieve at par or superior performance and stability in reinforcement learning tasks.

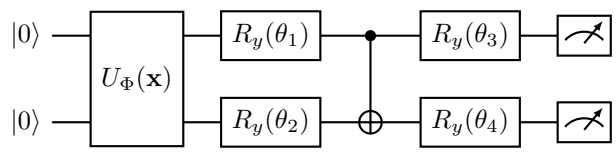

Fig. 1. Two-qubit variational circuit example.

II. Related Works

A. Foundations and Manual Architectures in Quantum Reinforcement Learning

The development of QRL has primarily centered on integrating VQCs into classical RL pipelines. Early work established that hybrid quantum–classical models can solve sequential decision-making tasks using significantly fewer trainable parameters than deep neural networks. Dragan *et al.* demonstrated a hybrid model for a stochastic slippery FrozenLake environment and used benchmarking circuits proposed by Sim *et al.* to study how gate type and connectivity affect performance.

Despite promising results, Dragan *et al.* observed that typical quantum metrics (expressibility, entanglement capability, and effective dimension) do not consistently predict

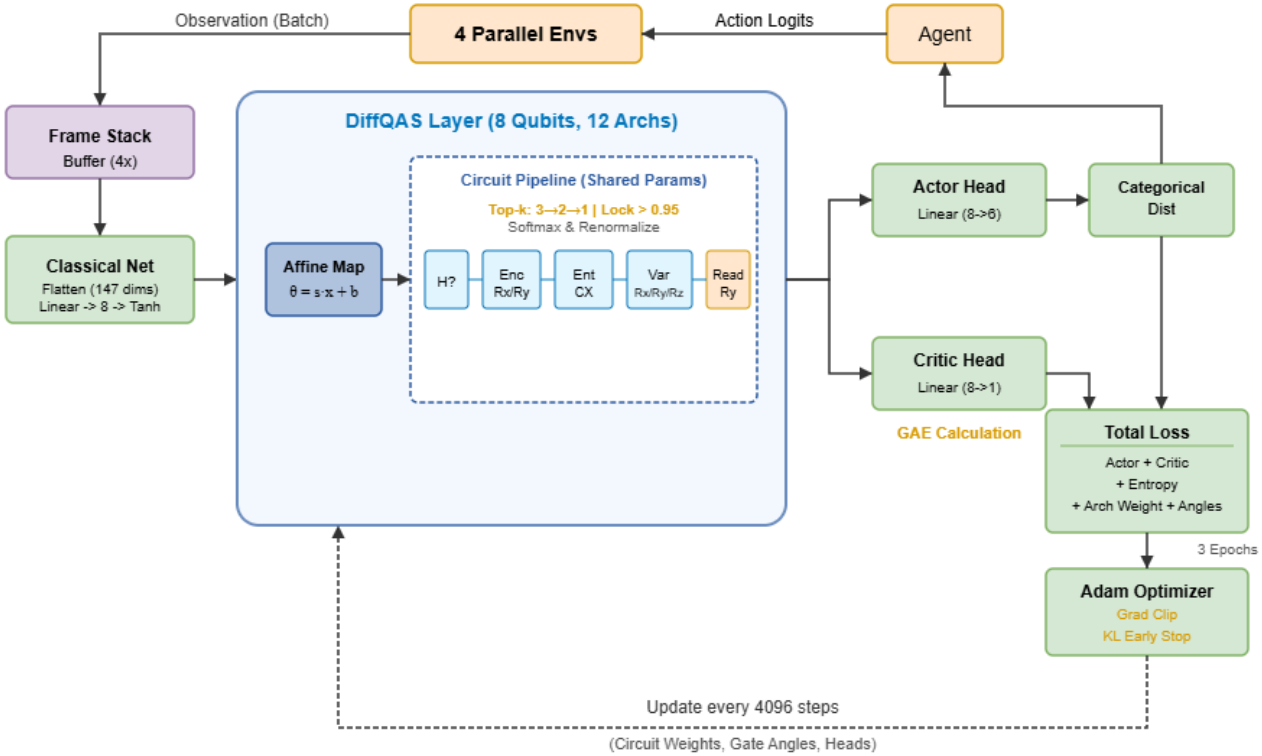

Fig. 2. Overview of the proposed DQACS framework.

QRL performance, suggesting that factors such as entangling-gate placement and interactions with classical post-processing layers may be crucial. As a result, constructing effective QRL models has traditionally required substantial domain expertise to manually navigate the combinatorial circuit design space.

B. Automated and Differentiable Quantum Architecture Search

To overcome manual design limitations, recent work has shifted toward automating circuit architecture engineering. Zhang *et al.* introduced DQAS by relaxing a discrete gate search space into a continuous, differentiable formulation, enabling gradient-based optimization of both circuit structure and variational parameters. Wu *et al.* extended this idea via QuantumDARTS, targeting noise-resilient architectures suitable for the NISQ era.

More recent advancements integrate differentiable search into RL training for improved efficiency and stability. Chen proposed asynchronous training to exploit parallelism during architecture search. Kumar *et al.* introduced RhoDARTS using density matrix simulations to incorporate noise models during search, improving robustness for realistic hardware. These works motivate a task-agnostic approach that can discover high-performing VQC architectures across diverse environments.

III. Methodology

A. DQAS Mechanism

The core of the model is a DQAS mechanism designed to find the circuit architecture best suited for the RL task. The DQAS circuit is 8 qubits wide and comprises 12 candidate architectures. For the encoding layer, the model chooses whether to include an initial Hadamard layer to create superposition and then selects between R_x or R_y encoding rotations that represent classical inputs as angles. For the ansatz, the model uses a chain entanglement pattern via controlled-NOT (CX) gates, followed by a final single-qubit rotation chosen from R_x, R_y, or R_z.

Rather than selecting components discretely during training, the method constructs 12 circuits corresponding to all combinations and uses a softmax distribution over structural weights to enable differentiation. To speed up search, certain parameters are shared across candidates: (i) the rotation angles for trainable gates, (ii) encoding affine-map parameters, and (iii) readout-rotation parameters.

Encoding is optimized by applying an affine map to every input feature,

$$\theta = \alpha\, x + \beta, \tag{1}$$

where α (`enc_scale`) and β (`enc_bias`) are shared trainable parameters. This constrains effective rotation angles to useful ranges (noting that values beyond 2π wrap around), improving encoding efficiency.

Readout is optimized by applying a shared R_y rotation on each qubit prior to measurement, providing a clearer signal for the readout stage.

B. Training with PPO

The overall system is a hybrid quantum–classical PPO agent. Each forward pass processes the environment observation using framestacking (current frame plus three previous frames) to capture motion and recent context. The stacked observation is flattened and compressed into 8 classical features, passed through a $\tanh$ activation to keep values in a range suitable for quantum rotations, and then fed into the quantum circuit(s).

To reduce compute during search, a top-k selection strategy is used: $k = 3$ for the first 20 updates, $k = 2$ until update 100, and then $k = 1$ to train only the highest-probability architecture. When an architecture weight exceeds 0.95, the model hard-selects a single circuit and skips further DQAS mixing. Structural weights are renormalized over the selected candidates so probabilities sum to 1.

The weighted outputs from the quantum architectures are passed to actor and critic heads to produce action logits and value estimates. During rollout, the agent interacts with four parallel environments and stores transitions in PPO memory. Once 4096 steps are collected, an update is triggered using Generalized Advantage Estimation (GAE) and three training epochs. The total loss combines architecture-weight loss, quantum gate-angle loss, entropy regularization, and standard actor–critic losses. To prevent policy collapse, KL early stopping and PPO gradient clipping are applied.

IV. Experimental Setup

To evaluate the efficacy of the proposed Differentiable Quantum Architecture Co-Search (DQACS) framework, we selected a diverse set of discrete-state reinforcement learning environments to thoroughly test the agent's navigation and sequential decision-making capabilities. We utilized three distinct configurations from the MiniGrid suite [4]. The MiniGrid-Empty-6x6 and MiniGrid-Empty-8x8 environments consist of unobstructed rooms where the agent must simply locate and navigate to a goal tile.

To test the agent against environmental obstacles, we also evaluated the framework on MiniGrid-SimpleCrossing-S9N1, which introduces walls and dead ends that require more complex pathfinding strategies. In these environments, the agent receives a compact vector representation of the grid and operates within a discrete action space. Furthermore, we trained the DQACS framework on the classic Frozen Lake 4x4 grid. This environment tasks the agent with navigating from a starting tile to a goal tile while avoiding hazardous hole tiles. Stepping on a hole immediately terminates the episode, forcing the agent to learn strictly safe traversal paths rather than purely optimizing for distance.

Sparse reward structures can significantly hinder the learning process of reinforcement learning agents by failing to provide intermediate guidance. To mitigate this in our more complex environments, specifically the Empty-8x8 and SimpleCrossing-S9N1, we implemented a custom reward shaping strategy. The agent is penalized with a -0.01 reward for every step taken to discourage inefficient wandering, and it receives a substantial +10 reward upon successfully reaching the goal to strongly reinforce task completion.

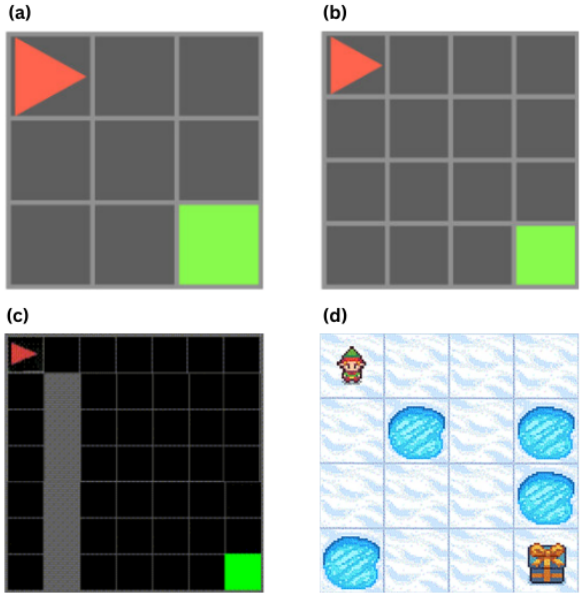

Fig. 3. Environment screenshots used for evaluation.

The DQACS framework was constructed as an end-to-end trainable hybrid quantum-classical model. The quantum circuit simulations, including the optimization of both the conventional circuit parameters and the structural architecture weights, were executed using PennyLane, an open-source framework for differentiable quantum programming. To handle the computational overhead associated with simulating parameterized quantum circuits and calculating policy gradients, the training pipeline was heavily parallelized and executed on a 16-core CPU cluster. This setup enables efficient asynchronous training across multiple environment instances, ensuring stable and robust policy updates.

V. Results

The empirical evaluation of the Differentiable Quantum Architecture Co-Search (DQACS) framework demonstrates robust learning capabilities across environments of varying complexity, effectively balancing the dual objectives of architectural optimization and policy learning. As illustrated in the performance metrics, the agent successfully solves all four tested environments, consistently achieving mean episode returns exceeding 0.9. In the stochastic Frozen Lake 4x4 environment, the framework exhibits rapid convergence, effectively solving the environment within approximately 200,000 global steps. The steep ascent of the reward curve indicates that the co-search mechanism quickly identified a quantum circuit structure capable of handling the stochastic transitions inherent

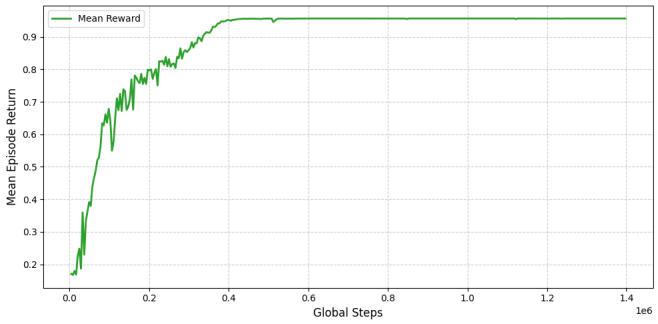

Fig. 4. Steps vs. mean reward MiniGrid-Empty-6x6.

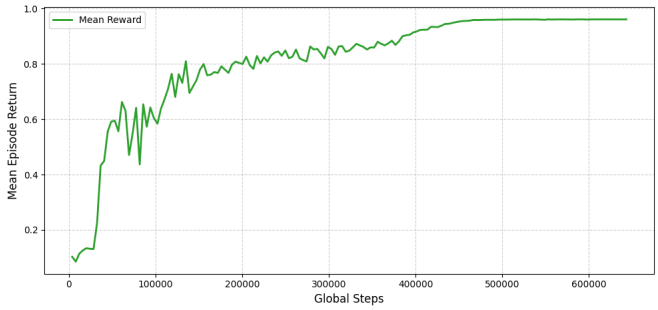

Fig. 5. Steps vs. mean reward MiniGrid-Empty-8x8.

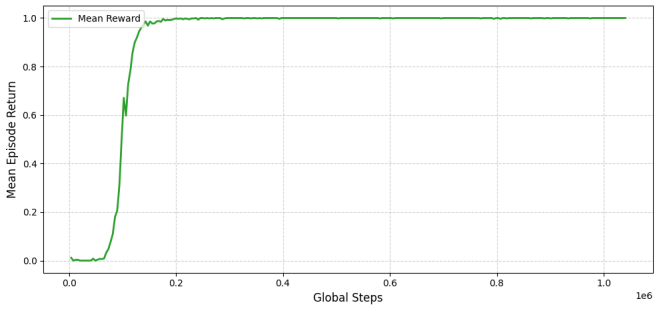

Fig. 6. Steps vs. mean reward FrozenLake-4x4.

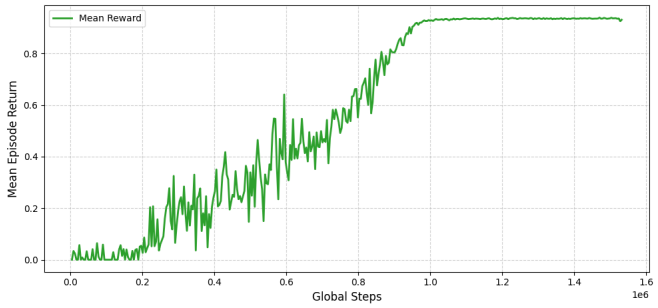

Fig. 7. Steps vs. mean reward MiniGrid-SimpleCrossing-S9N1.

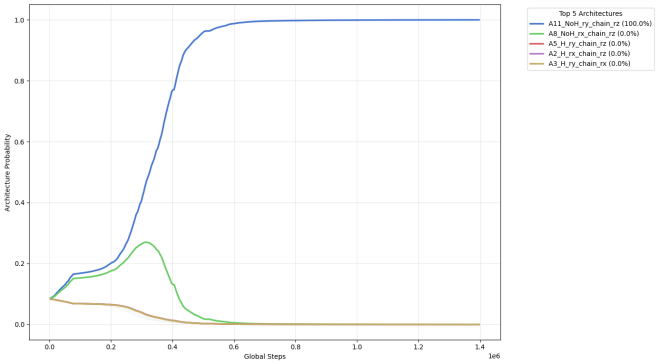

Fig. 8. Steps vs. architecture probability MiniGrid-Empty-6x6.

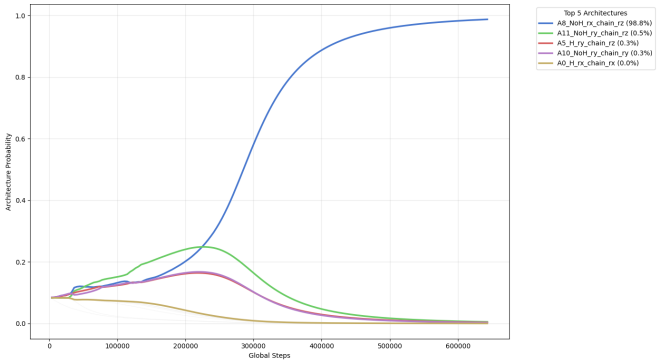

Fig. 9. Steps vs. architecture probability MiniGrid-Empty-8x8.

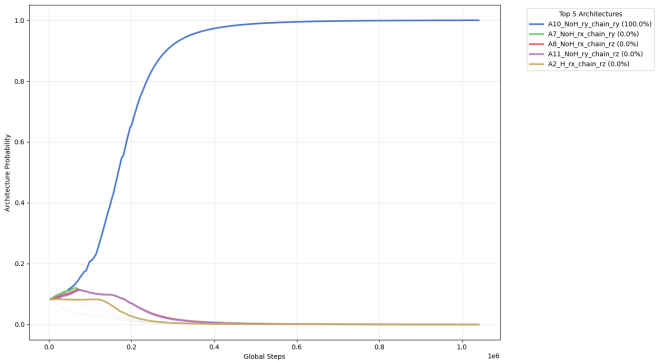

Fig. 10. Steps vs. architecture probability FrozenLake-4x4.

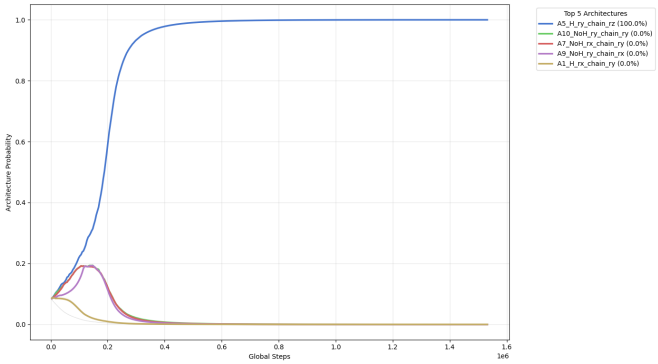

Fig. 11. Steps vs. architecture probability MiniGrid-SimpleCrossing-S9N1.

to the slippery grid, maintaining a near-perfect mean reward of 1.0 for the remainder of the training duration. This stability aligns with findings in prior quantum reinforcement learning (QRL) studies, which suggest that once an optimal ansatz is

located, hybrid models can exhibit lower variance in asymptotic performance compared to purely classical counterparts.

The scalability of the DQACS framework is further evidenced by its performance on the MiniGrid-Empty-6x6 and MiniGrid-Empty-8x8 tasks. In the 6x6 configuration, the agent demonstrates a smooth learning curve, overcoming the initial exploration phase to reach optimal performance stability around 400,000 steps. When scaled to the larger 8x8 grid, the increased state space introduces a noticeable "warm-up" period during the first 50,000 to 100,000 steps, where the mean reward remains near zero. This delay is characteristic of sparse reward environments where the agent must engage in extensive stochastic exploration and wandering before stumbling upon the objective.

However, once the policy gradient begins to propagate valid reward signals, the system rapidly improves, converging to a mean return of approximately 0.95 by 500,000 steps. This behavior highlights the ability of the differentiable search to adapt its circuit complexity to larger input dimensions without succumbing to the barren plateaus often observed in fixed-structure variational circuits.

The MiniGrid-SimpleCrossing-S9N1 environment presented the most significant challenge, requiring the agent to navigate around obstacles and dead ends. As reflected in the training trajectory, the agent effectively required 200,000 steps of exploration before achieving consistent positive reinforcement. Unlike the empty grids, the learning curve here exhibits a more gradual, linear ascent, fully converging only after approximately 1.0 million steps. This extended training horizon is consistent with the increased difficulty of the pathfinding task, yet the final asymptotic performance stabilizes above 0.9, validating the efficacy of the custom reward shaping implemented in the setup.

Furthermore, the correlation between reward stabilization and architecture probability is critical; as the differentiable search weights consolidate around a single "top" architecture, the variance in the episode return diminishes significantly. This suggests that the DQACS framework moves from a diverse super-net exploration phase to a focused exploitation phase, effectively "locking in" the optimal quantum circuit structure once the task is learned.

VI. Conclusion

This paper proposes Differentiable Quantum Architecture Co-Search (DQACS), extending differentiable quantum architecture search to jointly optimize encoding, ansatz, and readout within a hybrid PPO training loop. By introducing trainable affine encoding maps, trainable readout rotations, and framestacking, the approach achieves stable learning across multiple environments. Future work includes scaling to larger environments and evaluating performance on quantum hardware or larger compute clusters.

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